Binary mixtures benchmark: system selection report

Generated 2026-10-20 by `select_systems.py`.

Hard filters

A system must pass every filter below to be a candidate. The filter names match the labels in the funnel table and in the `failures` column of `candidates.csv`. Water mixtures can be admitted as listed exceptions to the composition-match and Sage repeat-SD filters (see Selected systems).

Filter	Rule	Why
(entry) ≥ 3 ΔHmix and ≥ 3 density points	At 297.0–299.5 K, counted per distinct composition.	Three compositions per system need a measured ΔHmix and density.
composition mismatch	Every ΔHmix composition has a density point within |Δx| ≤ 0.05.	Both properties are scored on the same box.
missing experimental uncertainty	Every ΔHmix and density point reports an uncertainty.	Needed for the MAZE denominator and to judge agreement.
relative ΔHmix uncertainty > 15 %	Largest relative ΔHmix uncertainty ≤ 15 %.	Errors in the passing pool are 0.5–3.5 %. Larger values are flat 25 % placeholders (e.g. 10.1016/j.jct.2013.09.015) or H^E near zero; either way they would inflate the MAZE denominator.
missing pure density with uncertainty	Both pure liquids have a density with an uncertainty in the window.	Needed for V^E and for scoring new pure densities.
elements	Only C, H, N, O and Cl.	Chosen scope: common organic and biomolecular elements.
too many heavy atoms	≤ 10 heavy atoms per molecule.	Keeps boxes small at MLIP cost while holding enough molecules.
viscous	No diol, polyol, polyamine or bis-hydroxyethyl amine motifs (SMARTS in `VISCOUS_SMARTS`).	Slow dynamics would not converge in 2 ns.
two-phase (linear H^E)	ΔHmix is not consistent with a straight line in x (three points: consecutive slopes differ by > 2σ; more points: reduced χ^2 of a linear fit > 2). σ has a 2 % relative floor. Applied only when the largest measured |H^E| ≥ 0.4 kJ/mol.	A two-phase region gives H^E = a + b·x (lever rule). An ideal or near-ideal mixture is also a straight line (H^E ≈ 0), so the test is skipped for small H^E; the small-signal filter handles those.
two-phase (H^E > RT/2)	ΔHmix(x≈0.5) ≤ RT/2 = 1.24 kJ/mol.	Regular-solution critical value; a conservative screen for an upper critical solution temperature above 298 K.
Sage repeat SD (hmix / density)	Largest published Sage 3-replicate SD over openff-2.0.0 to 2.3.0rc1 ≤ 0.1 kJ/mol (ΔHmix) and ≤ 0.001 g/mL (density).	A large spread between replicates suggests slow equilibration.
no Sage repeat SD (hmix / density)	A published Sage SD exists for the property.	Without it the convergence check above cannot be made.
small signal	|ΔHmix(x≈0.5)| ≥ 0.4 kJ/mol.	Expected simulation σ is 0.15–0.25 kJ/mol, so smaller values cannot separate models.

Systems are also checked for literature evidence of miscibility at the run temperature (see Per-system details); that check is manual.

Data sources

- OpenFF Sage 2.0 training set (openff-sage@a6a061b): no experimental uncertainties.
- Sage 2.3 refit training and validation sets (ash-sage-rc2@00c9524): experimental uncertainties for every point.
- Published Sage benchmark results (ash-sage-rc2 `04_benchmark`): mean and SD over 3 independent replicates for openff-2.0.0, 2.1.0, 2.2.1 and 2.3.0rc1.
- Full isotherms and pure densities from the NIST TRC ThermoML archive, fetched by DOI. ThermoML stores either a standard uncertainty (one standard deviation) or an expanded uncertainty (a 95 % interval, coverage factor k = 2). Standard uncertainties are used as given; expanded ones are divided by 2. Every σ in this report

is therefore a standard uncertainty. The ash-sage-rc2 sets use the same convention: their σ equals half the ThermoML expanded value.
- ml-peg's liquid_densities inputs, to avoid re-scoring pure densities.

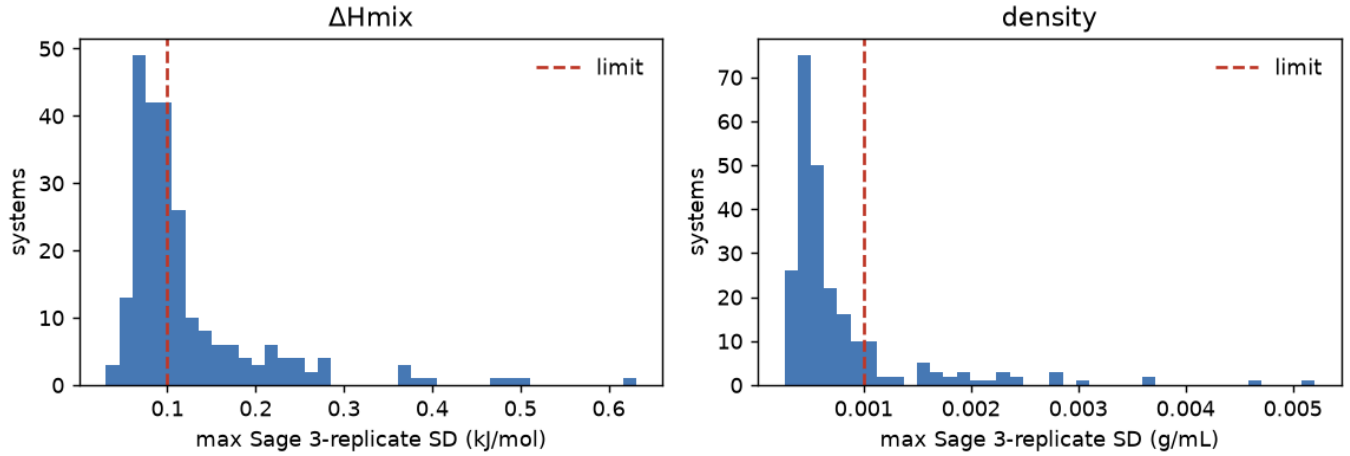
Screening funnel

- Binary systems in the Sage sets: 370.
- With ≥ 3 ΔHmix and ≥ 3 density points at 297.0–299.5 K: 248.
- Passing every hard filter: 36.

Failures by filter (a system can fail several):

Filter	Systems failing
Sage repeat SD (hmix)	102
small signal	97
composition mismatch	81
elements	51
Sage repeat SD (density)	42
missing pure density with uncertainty	40
two-phase (H^E > RT/2)	21
too many heavy atoms	14
relative ΔHmix uncertainty > 15 %	10
missing experimental uncertainty	9
no Sage repeat SD (hmix)	7
no Sage repeat SD (density)	7
viscous	7
two-phase (linear H^E)	2

The H^E-linearity test flags systems whose enthalpies are consistent with a straight line in x (the lever-rule signature of a two-phase region, H^E = a + b·x) and whose largest |H^E| is at least 0.4 kJ/mol. Systems it flags in the pool, with the largest relative ΔHmix uncertainty: `O` + `O=C1CCCN1` (25 %), `CN1CCCC1=O` + `c1ccncc1` (25 %). Large relative uncertainties can hide real curvature, so a flag on data with errors of ~25 % (placeholder values in the rc2 sets) is inconclusive rather than evidence of two phases. Such systems are also rejected by the relative-uncertainty filter.



Recommended fast-converging systems

System	Points (ΔHmix/ρ)	Linear H^E (two-phase signature)	Slopes between points	Outcome
`CCO` + `O`	3/3	no	1.244, 0.597	excluded: Already covered by the ethanol_water_density benchmark.

System	Points (ΔHmix/ρ)	Linear H^E (two-phase signature)	Slopes between points	Outcome
`C1CCNCC1` + `O`	3/3	no	-0.695, 2.700	excluded: Dropped after the Sage reference run (2026-10-20). Nitrogen inversion is frozen in Sage (no axial/equatorial N-H interconversion in 2.2 ns, or in a 500 K anneal), and the real barrier (~25 kJ/mol) suggests ns lifetimes for MLIPs too, so ΔHmix depended on the starting N-H orientation by ~0.3 kJ/mol. It also showed the largest box-size gap against the published 1000-molecule Sage run (-0.35 kJ/mol at x = 0.25) and is microheterogeneous for x between 0.08 and 0.5 (arXiv:2203.01224).
`CO` + `O`	3/3	no	0.365, 0.970	selected
`O` + `O=C1CCCN1`	3/3	yes	-0.551, 1.277	excluded: Viscous lactam that melts at ~25 °C. The Sage data use the imidic acid tautomer (OC1=NCCC1), so there is no valid Sage repeat SD, and the experimental enthalpy uncertainties are a flat 25 % placeholder.
`O` + `c1ccncc1`	3/0	no	-0.766, 0.906	selected
`CC(=O)c1ccccc1` + `O`	3/0	yes	0.677, 0.671	excluded: two-phase. Aqueous solubility is about 6 g/L at 25 °C (x ≈ 0.001), and the H^E slopes between x = 0.2, 0.5 and 0.8 are identical (0.677 and 0.671 kJ/mol), the lever-rule signature. The water content of the organic-rich phase was not found; confirm against the IUPAC-NIST Solubility Data Series.
`CC(=O)CC(C)=O` + `O`	3/0	yes	0.840, 0.867	excluded: two-phase. Binary mutual solubilities in ThermoML (10.1016/j.fluid.2010.11.029): at 298.15 K the aqueous phase holds an acetylacetone mass fraction of 0.158 (x ≈ 0.033), and the organic phase holds 3–6 wt% water between 288 and 313 K (x(acetylacetone) ≈ 0.75–0.85). The measured compositions (x = 0.2–0.75) are therefore two-phase.
`CCOC(=O)CC(C)=O` + `O`	3/0	yes	1.710, 1.469	excluded: two-phase. Aqueous solubility is of order 3 g/100 mL near room temperature (x ≈ 0.004; handbook value, not yet verified against a primary source). H^E/x is roughly constant (slopes 1.71 and 1.47 kJ/mol); x = 0.75 may lie near the organic-rich phase boundary.

- **acetophenone + water**: Aqueous solubility is about 6 g/L at 25 °C (x ≈ 0.001), and the H^E slopes between x = 0.2, 0.5 and 0.8 are identical (0.677 and 0.671 kJ/mol), the lever-rule signature. The water content of the organic-rich phase was not found; confirm against the IUPAC-NIST Solubility Data Series.
- **acetylacetone + water**: Binary mutual solubilities in ThermoML (10.1016/j.fluid.2010.11.029): at 298.15 K the aqueous phase holds an acetylacetone mass fraction of 0.158 (x ≈ 0.033), and the organic phase holds 3–6 wt% water between 288 and 313 K (x(acetylacetone) ≈ 0.75–0.85). The measured compositions (x = 0.2–0.75) are therefore two-phase.
- **ethyl acetoacetate + water**: Aqueous solubility is of order 3 g/100 mL near room temperature (x ≈ 0.004; handbook value, not yet verified against a primary source). H^E/x is roughly constant (slopes 1.71 and 1.47 kJ/mol); x = 0.75 may lie near the organic-rich phase boundary.

Selected systems

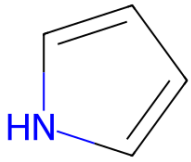
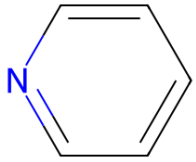
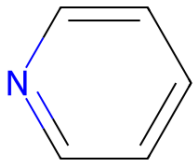
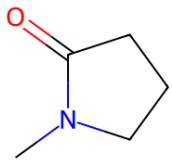
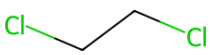
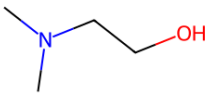
#	System	T (K)	ΔHmix at box x (kJ/mol)	V^E at box x (cm³/mol)	Hard filters	Interaction
1	pyrrole + pyridine	298.15	-1.77 / -2.65 / -1.62	-0.00 / 0.07 / 0.10	pass	N-H···N hydrogen bond between an aromatic donor and acceptor
2	methanol + water	298.15	-0.80 / -0.78 / -0.58	-0.62 / -1.00 / -0.81	exception	alcohol-water hydrogen-bond network
3	water + pyridine	303.15	-1.13 / -1.28 / -1.10	– / – / –	exception	aromatic nitrogen-water hydrogen bonding
4	N-methyl-2-pyrrolidone + 1,2-dichloroethane	298.15	-0.82 / -1.26 / -0.85	-0.09 / -0.12 / -0.08	exception	tertiary amide with a chloroalkane (C-H···O=C, dipolar)
5	2-(dimethylamino)ethanol + methanol	298.15	-0.80 / -0.87 / -0.53	-0.90 / -1.05 / -0.57	pass	amino-alcohol with an alcohol (O-H···N and O-H···O)
6	acetone + N-methylaniline	298.15	-0.50 / -0.67 / -0.53	-0.45 / -0.73 / -0.67	pass	aromatic N-H donor with a ketone acceptor
7	methyl pentanoate + octane	298.15	0.77 / 1.05 / 0.79	0.50 / 0.62 / 0.47	pass	dilution of ester dipoles in an alkane
8	1-propanol + benzene	298.15	1.04 / 0.97 / 0.55	0.14 / 0.09 / 0.02	pass	alcohol self-association broken by an aromatic (O-H···π)

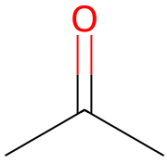
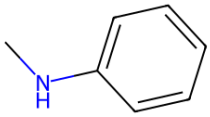
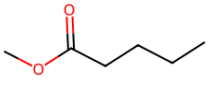
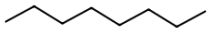
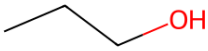
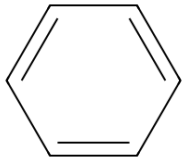
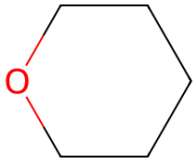
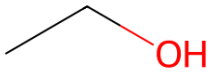
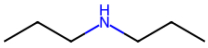
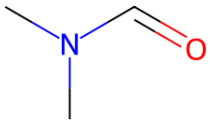
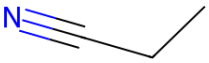
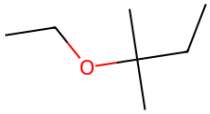
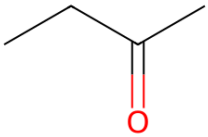
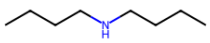
#	System	T (K)	ΔHmix at box x (kJ/mol)	V^E at box x (cm³/mol)	Hard filters	Interaction
9	tetrahydropyran + ethanol	298.15	0.57 / 0.85 / 0.75	-0.01 / 0.01 / 0.04	pass	cyclic ether with an alcohol
10	dipropylamine + N,N-dimethylformamide	298.15	0.61 / 0.75 / 0.53	-0.25 / -0.29 / -0.21	pass	secondary amine with a tertiary amide
11	propionitrile + tert-amyl ethyl ether	298.15	0.50 / 0.70 / 0.59	-0.22 / -0.37 / -0.33	pass	nitrile dipoles diluted in an ether
12	2-butanone + dibutylamine	298.15	0.42 / 0.62 / 0.43	0.17 / 0.26 / 0.20	pass	ketone with a secondary amine
13	heptane + toluene	298.15	0.45 / 0.56 / 0.39	0.11 / 0.15 / 0.11	pass	dispersion and π control (no polar groups)

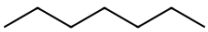
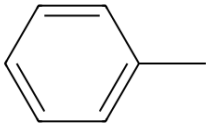
13 systems, 39 mixture boxes and 25 pure boxes (64 in total).

Structures

Component a is the one whose mole fraction x is quoted.

#	Component a	Component b
1	 pyrrole `c1cc[nH]c1`	 pyridine `c1ccncc1`
2	 methanol `CO`	 water `O`
3	 water `O`	 pyridine `c1ccncc1`
4	 N-methyl-2-pyrrolidone `CN1CCCC1=O`	 1,2-dichloroethane `ClCCCl`
5	 2-(dimethylamino)ethanol `CN(C)CCO`	 methanol `CO`

#	Component a	Component b
6	 acetone `CC(C)=O`	 N-methylaniline `CNc1ccccc1`
7	 methyl pentanoate `CCCCC(=O)OC`	 octane `CCCCCCCC`
8	 1-propanol `CCCO`	 benzene `c1ccccc1`
9	 tetrahydropyran `C1CCOCC1`	 ethanol `CCO`
10	 dipropylamine `CCCNCCC`	 N,N-dimethylformamide `CN(C)C=O`
11	 propionitrile `CCC#N`	 tert-amyl ethyl ether `CCOC(C)(C)CC`
12	 2-butanone `CCC(C)=O`	 dibutylamine `CCCCNCCCC`

#	Component a	Component b
13	 heptane `CCCCCCC`	 toluene `Cc1ccccc1`

V^E coverage

V^E diversity was not a selection criterion. Sign at the three box compositions:

- negative: 6
- positive: 4
- sign change: 2
- no density data: 1

Box plan

Boxes have edge ≥ 25.0 Å, ≥ 100 molecules and ≥ 20 minority molecules, with x within 0.01 of the measured composition. Evaluator, by comparison, uses a fixed total (1000 molecules for Sage), rounds x·N per component and only requires at least one molecule of each, so at the compositions used here its minority counts are about 200 or more.

Mixture boxes: 1144–2470 atoms (median 1495); total 62760 atoms across 39 boxes. Constraint that sets the size: edge 25, total 14.

System	x (box)	Molecules a / b	Total	Atoms	Edge (Å)	Set by
pyrrole + pyridine	0.240	29 / 92	121	1302	25.0	edge
pyrrole + pyridine	0.508	64 / 62	126	1322	25.1	edge
pyrrole + pyridine	0.763	100 / 31	131	1341	25.1	edge
methanol + water	0.199	86 / 346	432	1554	25.0	edge
methanol + water	0.500	168 / 168	336	1512	25.0	edge
methanol + water	0.701	202 / 86	288	1470	25.0	edge
water + pyridine	0.296	45 / 107	152	1312	25.1	edge
water + pyridine	0.495	94 / 96	190	1338	25.0	edge
water + pyridine	0.700	180 / 77	257	1387	25.0	edge
N-methyl-2-pyrrolidone + 1,2-dichloroethane	0.254	29 / 85	114	1144	25.0	edge
N-methyl-2-pyrrolidone + 1,2-dichloroethane	0.500	54 / 54	108	1296	25.1	edge
N-methyl-2-pyrrolidone + 1,2-dichloroethane	0.740	77 / 27	104	1448	25.1	edge
2-(dimethylamino)ethanol + methanol	0.247	41 / 125	166	1447	25.0	edge
2-(dimethylamino)ethanol + methanol	0.500	65 / 65	130	1495	25.1	edge
2-(dimethylamino)ethanol + methanol	0.750	81 / 27	108	1539	25.0	edge
acetone + N-methylaniline	0.230	23 / 77	100	1539	25.6	total
acetone + N-methylaniline	0.505	51 / 50	101	1360	25.1	edge
acetone + N-methylaniline	0.709	78 / 32	110	1324	25.0	edge
methyl pentanoate + octane	0.230	23 / 77	100	2462	29.6	total
methyl pentanoate + octane	0.490	49 / 51	100	2306	29.1	total
methyl pentanoate + octane	0.730	73 / 27	100	2162	28.6	total
1-propanol + benzene	0.245	27 / 83	110	1320	25.1	edge
1-propanol + benzene	0.500	57 / 57	114	1368	25.1	edge
1-propanol + benzene	0.750	90 / 30	120	1440	25.0	edge
tetrahydropyran + ethanol	0.250	34 / 102	136	1462	25.0	edge
tetrahydropyran + ethanol	0.504	60 / 59	119	1491	25.0	edge

System	x (box)	Molecules a / b	Total	Atoms	Edge (Å)	Set by
tetrahydropyran + ethanol	0.755	80 / 26	106	1514	25.0	edge
dipropylamine + N,N-dimethylformamide	0.248	26 / 79	105	1520	25.1	edge
dipropylamine + N,N-dimethylformamide	0.500	50 / 50	100	1700	26.1	total
dipropylamine + N,N-dimethylformamide	0.750	75 / 25	100	1950	27.3	total
propionitrile + tert-amyl ethyl ether	0.200	20 / 80	100	2100	28.3	total
propionitrile + tert-amyl ethyl ether	0.500	50 / 50	100	1650	26.4	total
propionitrile + tert-amyl ethyl ether	0.700	70 / 30	100	1350	25.1	edge
2-butanone + dibutylamine	0.220	22 / 78	100	2470	29.4	total
2-butanone + dibutylamine	0.500	50 / 50	100	2050	27.9	total
2-butanone + dibutylamine	0.790	79 / 21	100	1615	26.1	total
heptane + toluene	0.250	25 / 75	100	1700	26.9	total
heptane + toluene	0.500	50 / 50	100	1900	27.6	total
heptane + toluene	0.750	75 / 25	100	2100	28.4	total

Sage baseline (published, 1000-molecule boxes)

System	Max rep. SD ΔHmix	Max rep. SD ρ	openff-2.0.0 ΔHmix MAE	openff-2.1.0 ΔHmix MAE	openff-2.2.1 ΔHmix MAE	openff-2.3.0rc1 ΔHmix MAE	openff-2.2.1 ρ MAE
pyrrole + pyridine	0.078	0.0005	1.905	1.922	1.930	2.023	0.0053
methanol + water	0.030	0.0010	0.815	0.826	0.825	0.833	0.0396
water + pyridine	–	–	–	–	–	–	–
N-methyl-2-pyrrolidone + 1,2-dichloroethane	0.062	0.0008	0.520	0.520	0.454	0.577	0.0216
2-(dimethylamino)ethanol + methanol	0.075	0.0005	0.069	0.326	0.393	0.171	0.0087
acetone + N-methylaniline	0.091	0.0004	0.024	0.126	0.172	0.088	0.0055
methyl pentanoate + octane	0.083	0.0004	0.366	0.275	0.277	0.274	0.0073
1-propanol + benzene	0.075	0.0004	0.212	0.195	0.227	0.292	0.0131
tetrahydropyran + ethanol	0.077	0.0005	0.047	0.096	0.084	0.154	0.0089
dipropylamine + N,N-dimethylformamide	0.069	0.0004	0.375	0.483	0.473	0.407	0.0087
propionitrile + tert-amyl ethyl ether	0.061	0.0003	0.286	0.355	0.347	0.382	0.0003
2-butanone + dibutylamine	0.093	0.0004	0.281	0.246	0.286	0.283	0.0097
heptane + toluene	0.076	0.0006	0.137	0.124	0.140	0.168	0.0163

Pure densities

Each pure density is taken from the paper that supplies most of the mixture densities, so V^E is self-consistent, unless an explicit reference is given (water from IAPWS-95). Other sources disagreeing by more than 0.002 g/mL are listed.

Component	T (K)	ρ (g/mL)	σ	Source	In ml-peg	ml-peg value	Disagreeing sources
`c1cc[nH]c1`	298.15	0.96554	0.00006	thermoml:10.1016/j.fluid.2013.10.029	no	–	–
`c1ccncc1`	298.15	0.97801	0.00006	thermoml:10.1016/j.fluid.2013.10.029	yes	0.9819 @ 298 K	thermoml:10.1016/j.fluid.2015.04.031, sage-2.0-train:10.1016/j.fluid.2015.04.031, ml-peg
`CO`	298.15	0.78657	0.00012	thermoml:10.1021/je700300y	yes	0.7914 @ 298 K	ml-peg

Component	T (K)	ρ (g/mL)	σ	Source	In ml-peg	ml-peg value	Disagreeing sources
`O`	298.15	0.99705	0.00000	IAPWS-95 (Wagner and Pruss, J. Phys. Chem. Ref. Data 31, 387 (2002))	yes	0.9980 @ 298 K	–
`O`	303.15	0.99565	0.00000	IAPWS-95 (Wagner and Pruss, J. Phys. Chem. Ref. Data 31, 387 (2002))	yes	0.9980 @ 298 K	–
`c1ccncc1`	303.15	0.97298	0.00011	thermoml:10.1021/je900874z	yes	0.9819 @ 298 K	–
`CN1CCCC1=O`	298.15	1.02800	0.00015	thermoml:10.1021/je7003758	yes	1.0230 @ 303 K	thermoml:10.1016/j.fluid.2008.02.023, rc2-validation:10.1016/j.fluid.2008.02.023
`ClCCCl`	298.15	1.24550	0.00030	thermoml:10.1021/je7003758	yes	1.2454 @ 303 K	–
`CN(C)CCO`	298.15	0.88334	0.00020	thermoml:10.1016/j.tca.2014.03.038	no	–	–
`CC(C)=O`	298.15	0.78532	0.00025	thermoml:10.1021/je900874z	yes	0.7845 @ 303 K	–
`CNc1ccccc1`	298.15	0.98207	0.00047	thermoml:10.1021/je900874z	no	–	–
`CCCCC(=O)OC`	298.15	0.88484	0.00050	thermoml:10.1021/acs.jced.5b00813	no	–	–
`CCCCCCCC`	298.15	0.69858	0.00052	thermoml:10.1021/acs.jced.5b00813	no	–	–
`CCCO`	298.15	0.79969	0.00013	thermoml:10.1021/je700254t	yes	0.7997 @ 303 K	thermoml:10.1021/je900212d, rc2-validation:10.1021/je900212d
`c1ccccc1`	298.15	0.87358	0.00008	thermoml:10.1021/je700254t	yes	0.8765 @ 298 K	ml-peg
`C1CCOCC1`	298.15	0.87903	0.00004	thermoml:10.1021/je201140q	no	–	–
`CCO`	298.15	0.78511	0.00006	thermoml:10.1021/je201140q	yes	0.7893 @ 298 K	ml-peg
`CCCNCCC`	298.15	0.73362	0.00089	thermoml:10.1021/acs.jced.5b00802	no	–	–
`CN(C)C=O`	298.15	0.94408	0.00113	thermoml:10.1021/acs.jced.5b00802	yes	0.9445 @ 303 K	thermoml:10.1007/s10765-016-2096-3, rc2-validation:10.1007/s10765-016-2096-3
`CCC#N`	298.15	0.77600	0.00015	thermoml:10.1021/je301238f	no	–	–
`CCOC(C)(C)CC`	298.15	0.76100	0.00015	thermoml:10.1021/je301238f	no	–	–
`CCC(C)=O`	298.15	0.80008	0.00009	thermoml:10.1016/j.tca.2010.09.004	yes	0.7999 @ 303 K	–
`CCCCNCCCC`	298.15	0.75546	0.00010	thermoml:10.1016/j.tca.2010.09.004	no	–	thermoml:10.1021/je800026p, rc2-validation:10.1021/je800026p
`CCCCCCC`	298.15	0.67954	0.00013	thermoml:10.1021/je400184t	no	–	–
`Cc1ccccc1`	298.15	0.86222	0.00011	thermoml:10.1021/je400184t	yes	0.8668 @ 298 K	ml-peg

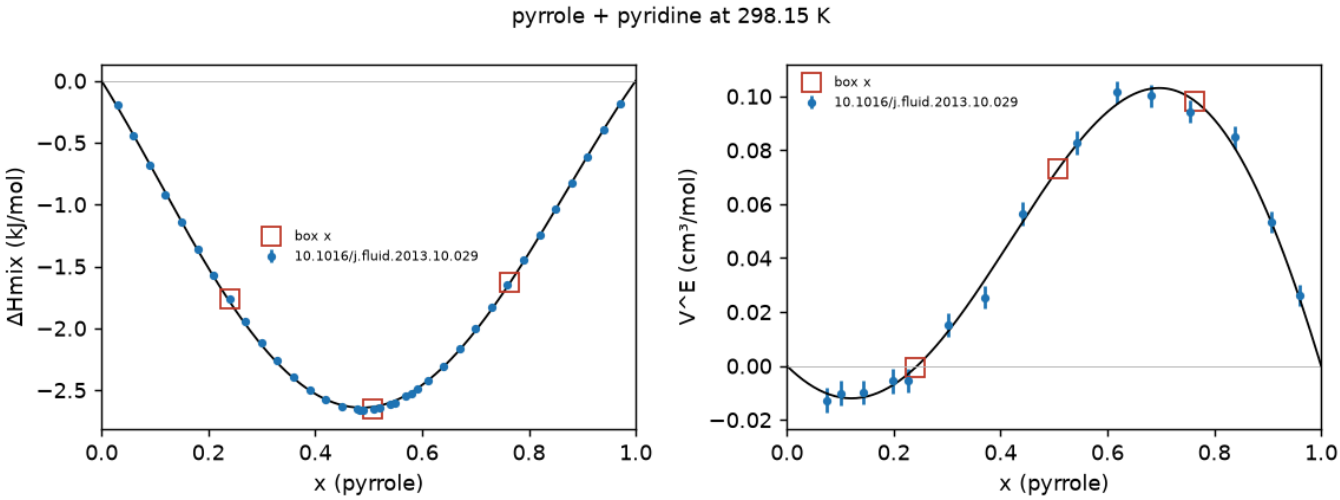
Many ml-peg liquid_densities references appear to be measured about 5 K below their labelled temperature (20 °C values labelled 298 K, 25 °C values labelled 303 K), so they are not used here; see `generation/data_issues.md`. Pure densities of components already in ml-peg are simulated (needed for ΔHmix) but not re-scored.

Per-system details

1. pyrrole + pyridine (`c1cc[nH]c1` + `c1ccncc1`)

- **Interaction:** N-H···N hydrogen bond between an aromatic donor and acceptor
- **Rationale:** Largest exothermic signal in the pool (-2.65 kJ/mol), single source paper for both properties, and Sage gets it badly wrong (MAE 1.9 kJ/mol).
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 38 points from 10.1016/j.fluid.2013.10.029 (38)
- **Density data:** 15 points from 10.1016/j.fluid.2013.10.029 (15)
- **Single-phase evidence:** ΔHmix measured at 38 compositions spanning x = 0.03–0.97; straight-line fit reduced χ^2 = 918 (a two-phase region would give ≈ 1).

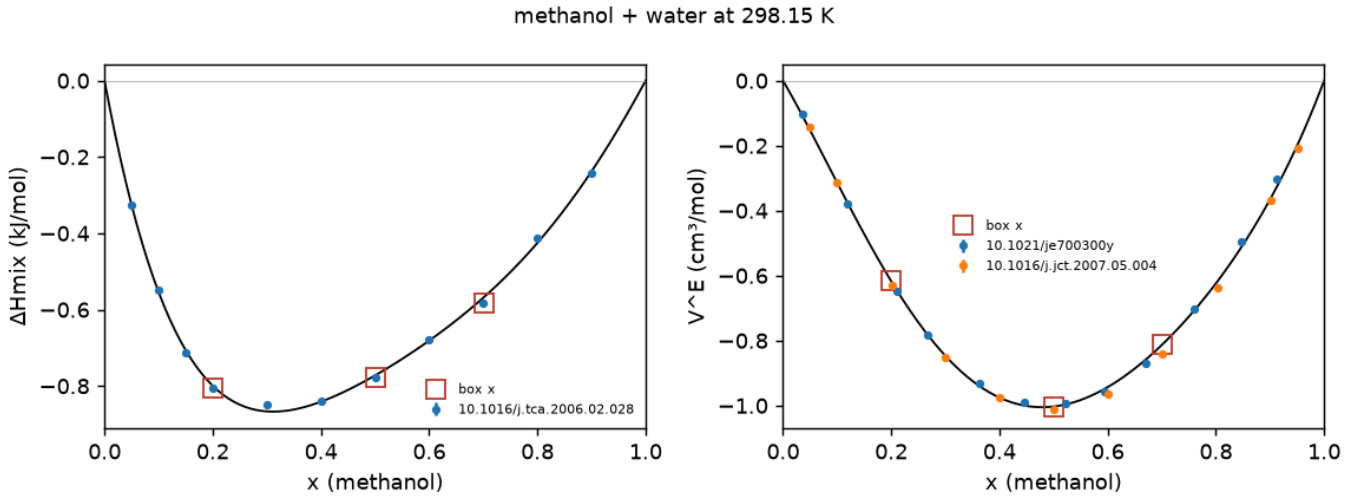
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.240	29 / 92	1302	25.0	-1.767 ± 0.0115	0.9754 ± 0.00006	-0.001
0.508	64 / 62	1322	25.1	-2.650 ± 0.0164	0.9712 ± 0.00006	0.073
0.763	100 / 31	1341	25.1	-1.625 ± 0.0107	0.9675 ± 0.00006	0.098



2. methanol + water (`CO` + `O`)

- **Interaction:** alcohol-water hydrogen-bond network
- **Rationale:** Smallest Sage repeat spread in the pool (0.03 kJ/mol), so it calibrates the noise of the protocol. Methanol and water pure densities are already scored in ml-peg and are not re-scored.
- **Miscibility:** Miscible in all proportions.
- **Filter exception:** Aqueous exception to rules 1 and 2. The rc2 density and enthalpy compositions differ by 0.06 in x (full ThermoML isotherms remove the mismatch), and water's pure density comes from IAPWS-95.
- **Failed filters:** composition mismatch; missing pure density with uncertainty
- **ΔHmix data:** 11 points from 10.1016/j.tca.2006.02.028 (11)
- **Density data:** 23 points from 10.1021/je700300y (12), 10.1016/j.jct.2007.05.004 (11)
- **Single-phase evidence:** ΔHmix measured at 11 compositions spanning x = 0.05–0.90; straight-line fit reduced χ^2 = 202 (a two-phase region would give ≈ 1).

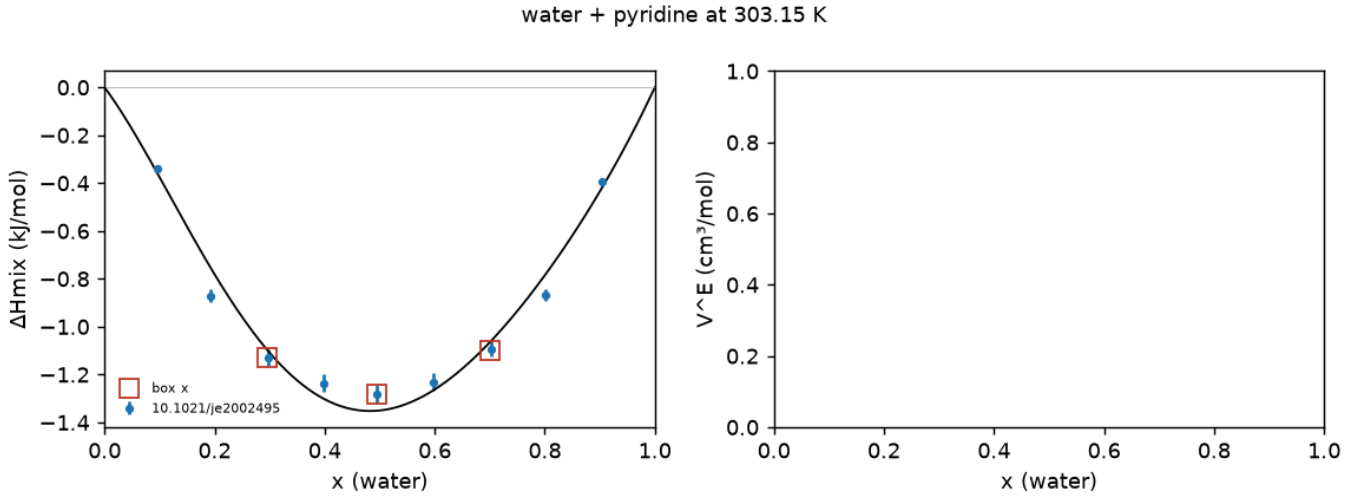
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.199	86 / 346	1554	25.0	-0.805 ± 0.0045	0.9473 ± 0.00006	-0.615
0.500	168 / 168	1512	25.0	-0.777 ± 0.0045	0.8813 ± 0.00009	-1.003
0.701	202 / 86	1470	25.0	-0.581 ± 0.0035	0.8401 ± 0.00010	-0.811



3. water + pyridine (`O` + `c1ccncc1`)

- **Interaction:** aromatic nitrogen-water hydrogen bonding
- **Rationale:** Only aromatic-water system available; relevant to biomolecular recognition.
- **Miscibility:** Miscible in all proportions at 303 K (enthalpies measured across x = 0.1–0.9).
- **Filter exception:** Aqueous exception: the only data are at 303.15 K and there are no published Sage repeat errors. Enthalpy only unless ThermoML provides a 303.15 K mixture density. MAZE uses our own Sage replicates.
- **ΔHmix data:** 9 points from 10.1021/je2002495 (9)
- **Density data:** 0 points from none
- **Single-phase evidence:** ΔHmix measured at 9 compositions spanning x = 0.10–0.91; straight-line fit reduced χ^2 = 344 (a two-phase region would give ≈ 1).

x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.296	45 / 107	1312	25.1	-1.127 ± 0.0341	– ± –	–
0.495	94 / 96	1338	25.0	-1.281 ± 0.0387	– ± –	–
0.700	180 / 77	1387	25.0	-1.097 ± 0.0331	– ± –	–

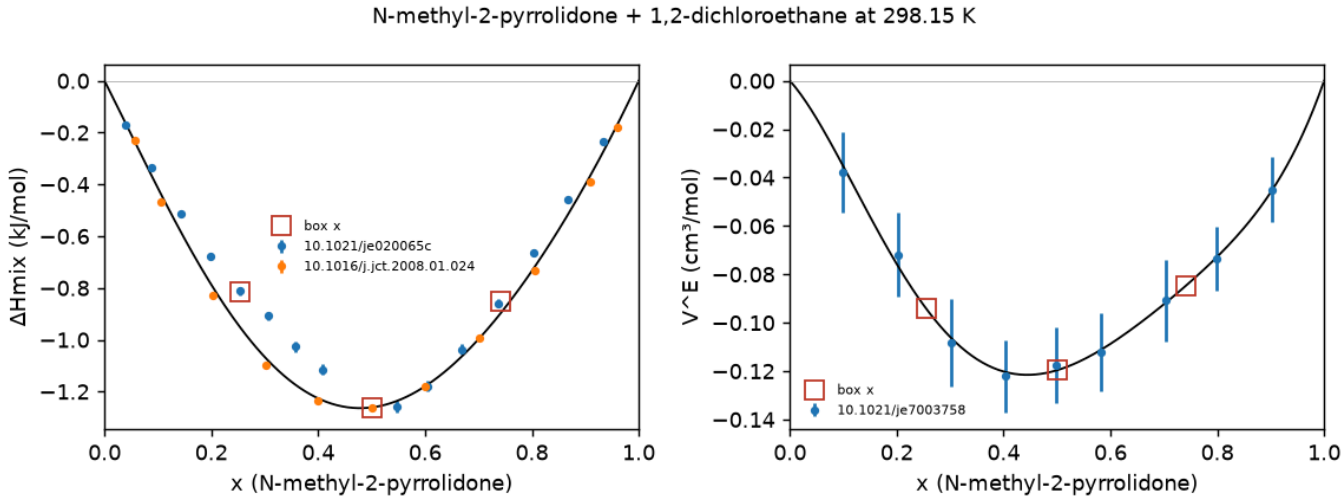


4. N-methyl-2-pyrrolidone + 1,2-dichloroethane (`CN1CCCC1=O` + `ClCCCl`)

- **Interaction:** tertiary amide with a chloroalkane (C-H···O=C, dipolar)
- **Rationale:** Widens the exothermic range (-1.26 kJ/mol) and covers chlorine. Both pure densities are already in ml-peg.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **Filter exception:** Fails rule 1 by a hair on the 3-point rc2 subsets (|dx| = 0.0503 vs 0.05). The full ThermoML isotherms remove the mismatch.
- **Failed filters:** composition mismatch

- **ΔHmix data:** 26 points from 10.1021/je020065c (15), 10.1016/j.jct.2008.01.024 (11)
- **Density data:** 9 points from 10.1021/je7003758 (9)
- **Single-phase evidence:** ΔHmix measured at 26 compositions spanning x = 0.04–0.96; straight-line fit reduced χ^2 = 714 (a two-phase region would give ≈ 1).

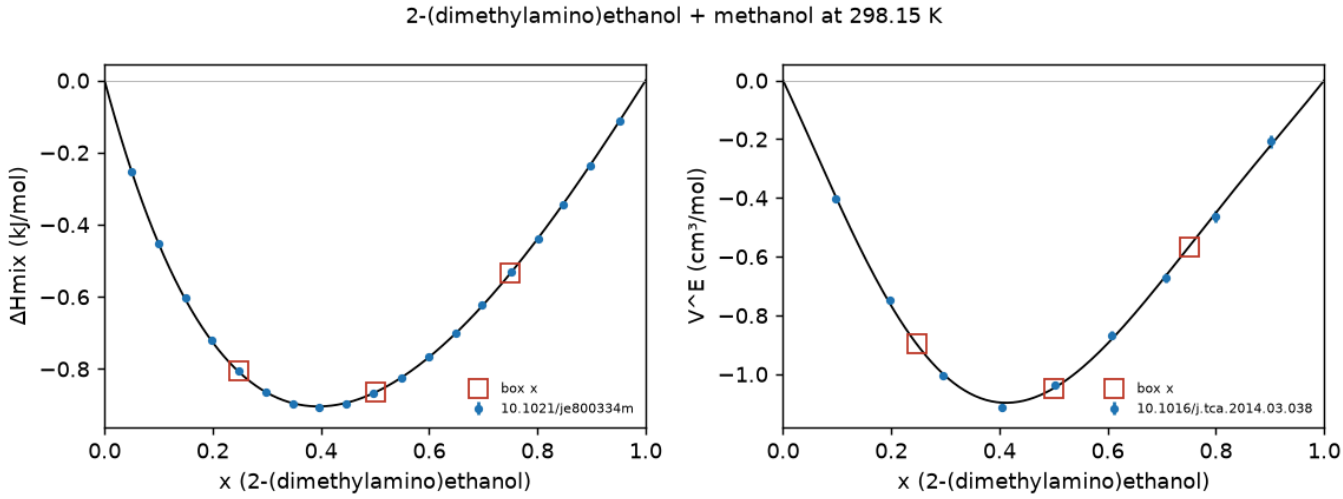
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.254	29 / 85	1144	25.0	-0.815 ± 0.0152	1.1831 ± 0.00025	-0.094
0.500	54 / 54	1296	25.1	-1.264 ± 0.0055	1.1278 ± 0.00020	-0.120
0.740	77 / 27	1448	25.1	-0.849 ± 0.0160	1.0777 ± 0.00020	-0.085



5. 2-(dimethylamino)ethanol + methanol (`CN(C)CCO` + `CO`)

- **Interaction:** amino-alcohol with an alcohol (O-H···N and O-H···O)
- **Rationale:** Exothermic cross-association between a bifunctional donor/acceptor and an alcohol.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 19 points from 10.1021/je800334m (19)
- **Density data:** 9 points from 10.1016/j.tca.2014.03.038 (9)
- **Single-phase evidence:** ΔHmix measured at 19 compositions spanning x = 0.05–0.95; straight-line fit reduced χ^2 = 291 (a two-phase region would give ≈ 1).

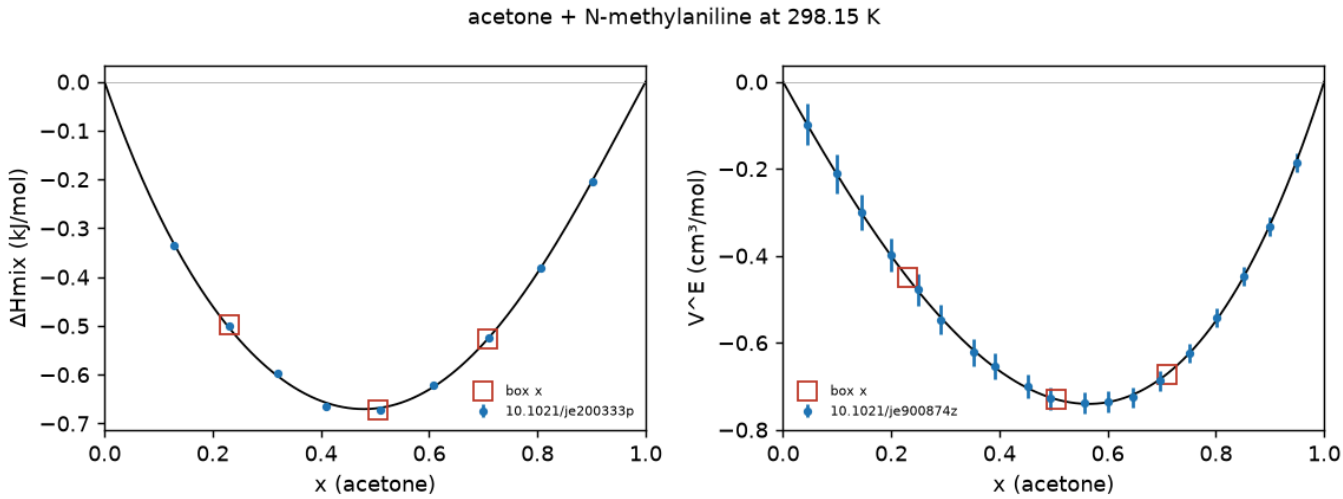
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.247	41 / 125	1447	25.0	-0.804 ± 0.0087	0.8437 ± 0.00014	-0.897
0.500	65 / 65	1495	25.1	-0.865 ± 0.0094	0.8684 ± 0.00015	-1.048
0.750	81 / 27	1539	25.0	-0.533 ± 0.0059	0.8777 ± 0.00017	-0.566



6. acetone + N-methylaniline (`CC(C)=O` + `CNc1ccccc1`)

- **Interaction:** aromatic N-H donor with a ketone acceptor
- **Rationale:** Exothermic N-H···O=C association; adds an aromatic amine.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 9 points from 10.1021/je200333p (9)
- **Density data:** 19 points from 10.1021/je900874z (19)
- **Single-phase evidence:** ΔHmix measured at 9 compositions spanning x = 0.13–0.90; straight-line fit reduced χ^2 = 393 (a two-phase region would give ≈ 1).

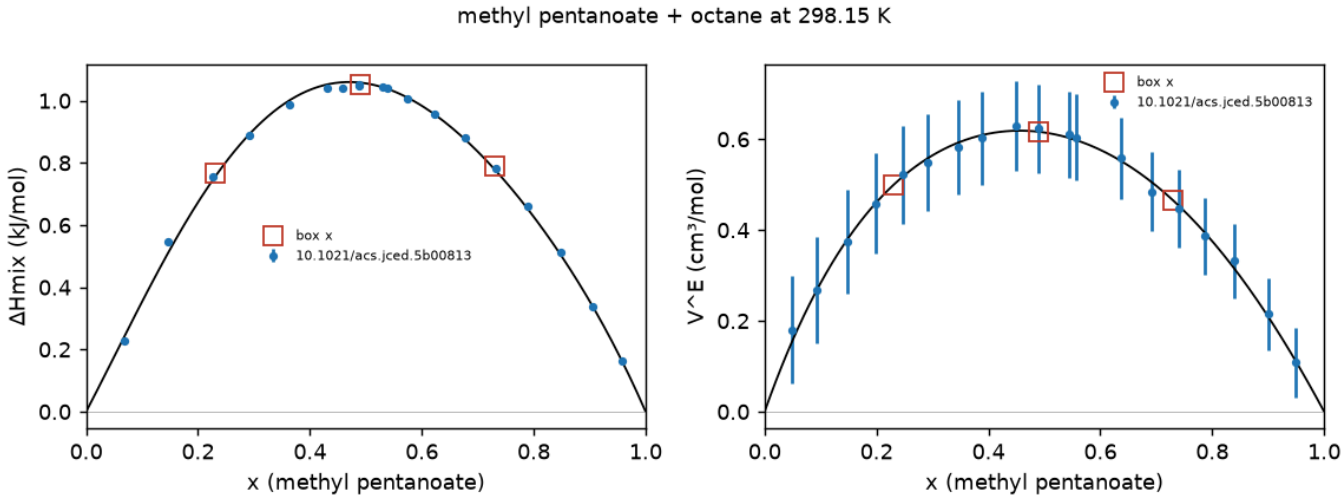
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.230	23 / 77	1539	25.6	-0.498 ± 0.0055	0.9532 ± 0.00035	-0.450
0.505	51 / 50	1360	25.1	-0.673 ± 0.0075	0.9089 ± 0.00026	-0.730
0.709	78 / 32	1324	25.0	-0.526 ± 0.0055	0.8664 ± 0.00023	-0.672



7. methyl pentanoate + octane (`CCCCC(=O)OC` + `CCCCCCCC`)

- **Interaction:** dilution of ester dipoles in an alkane
- **Rationale:** Endothermic dipole dilution, neither component in ml-peg.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below). Endothermic ester + alkane; check the demixing diagnostic in the pilot.
- **ΔHmix data:** 19 points from 10.1021/acs.jced.5b00813 (19)
- **Density data:** 19 points from 10.1021/acs.jced.5b00813 (19)
- **Single-phase evidence:** ΔHmix measured at 19 compositions spanning x = 0.07–0.96; straight-line fit reduced χ^2 = 656 (a two-phase region would give ≈ 1).

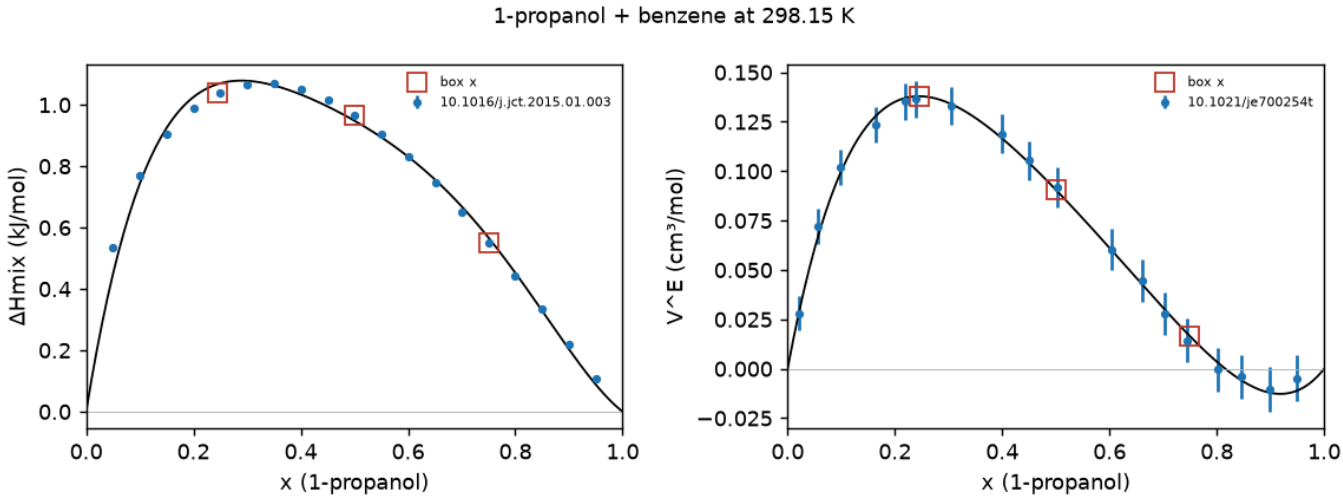
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.230	23 / 77	2462	29.6	0.769 ± 0.0080	0.7323 ± 0.00051	0.499
0.490	49 / 51	2306	29.1	1.053 ± 0.0115	0.7764 ± 0.00051	0.616
0.730	73 / 27	2162	28.6	0.789 ± 0.0085	0.8234 ± 0.00051	0.465



8. 1-propanol + benzene (`CCCO` + `c1ccccc1`)

- **Interaction:** alcohol self-association broken by an aromatic (O-H···π)
- **Rationale:** Alcohol dilution case that also probes O-H···π, replacing isooctane + 1-propanol. Both pure densities are already in ml-peg.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 19 points from 10.1016/j.jct.2015.01.003 (19)
- **Density data:** 18 points from 10.1021/je700254t (18)
- **Single-phase evidence:** ΔHmix measured at 19 compositions spanning x = 0.05–0.95; straight-line fit reduced χ^2 = 142 (a two-phase region would give ≈ 1).

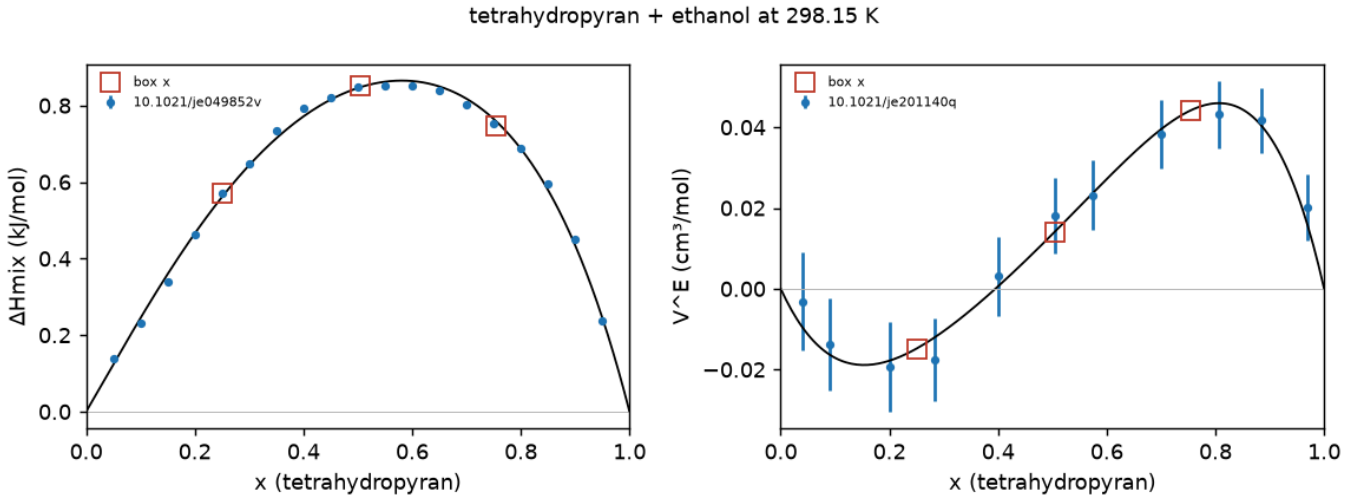
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.245	27 / 83	1320	25.1	1.039 ± 0.0065	0.8563 ± 0.00009	0.138
0.500	57 / 57	1368	25.1	0.968 ± 0.0065	0.8389 ± 0.00010	0.091
0.750	90 / 30	1440	25.0	0.551 ± 0.0060	0.8205 ± 0.00011	0.016



9. tetrahydropyran + ethanol (`C1CCOCC1` + `CCO`)

- **Interaction:** cyclic ether with an alcohol
- **Rationale:** Endothermic; ether acceptor competes with alcohol self-association.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 19 points from 10.1021/je049852v (19)
- **Density data:** 11 points from 10.1021/je201140q (11)
- **Single-phase evidence:** ΔHmix measured at 19 compositions spanning x = 0.05–0.95; straight-line fit reduced χ^2 = 374 (a two-phase region would give ≈ 1).

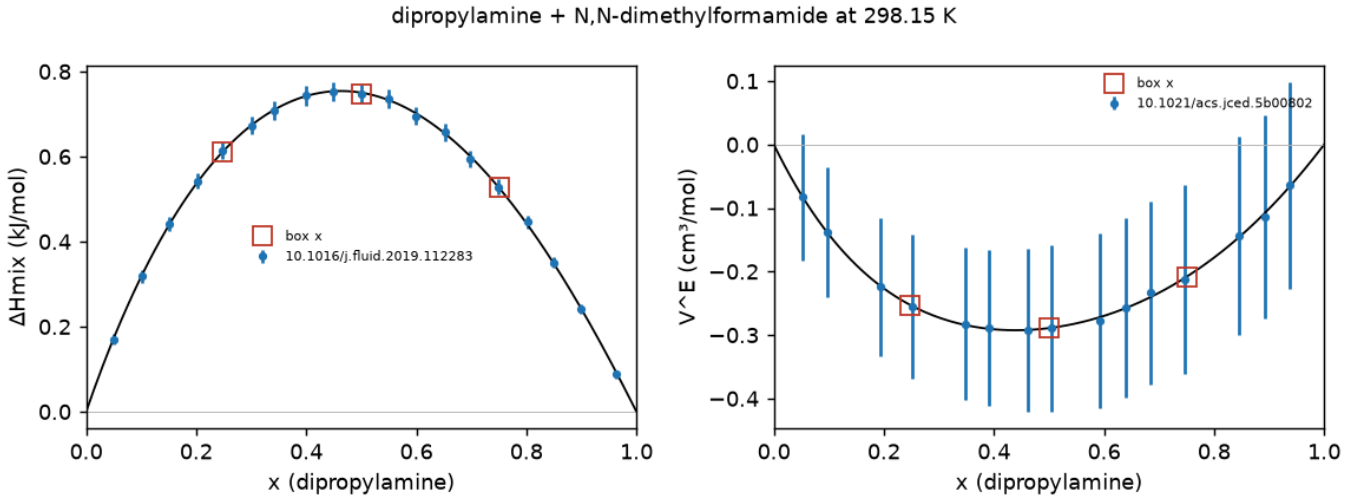
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.250	34 / 102	1462	25.0	0.570 ± 0.0055	0.8189 ± 0.00013	-0.015
0.504	60 / 59	1491	25.0	0.851 ± 0.0079	0.8441 ± 0.00010	0.014
0.755	80 / 26	1514	25.0	0.747 ± 0.0071	0.8633 ± 0.00008	0.044



10. dipropylamine + N,N-dimethylformamide (`CCCNCCC` + `CN(C)C=O`)

- **Interaction:** secondary amine with a tertiary amide
- **Rationale:** Endothermic amine/amide pair; dipropylamine is new to ml-peg.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 19 points from 10.1016/j.fluid.2019.112283 (19)
- **Density data:** 15 points from 10.1021/acs.jced.5b00802 (15)
- **Single-phase evidence:** ΔHmix measured at 19 compositions spanning x = 0.05–0.96; straight-line fit reduced $\chi^2 = 99$ (a two-phase region would give ≈ 1).

x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.248	26 / 79	1520	25.1	0.612 ± 0.0195	0.8687 ± 0.00106	-0.253
0.500	50 / 50	1700	26.1	0.748 ± 0.0225	0.8115 ± 0.00099	-0.289
0.750	75 / 25	1950	27.3	0.528 ± 0.0170	0.7681 ± 0.00094	-0.208

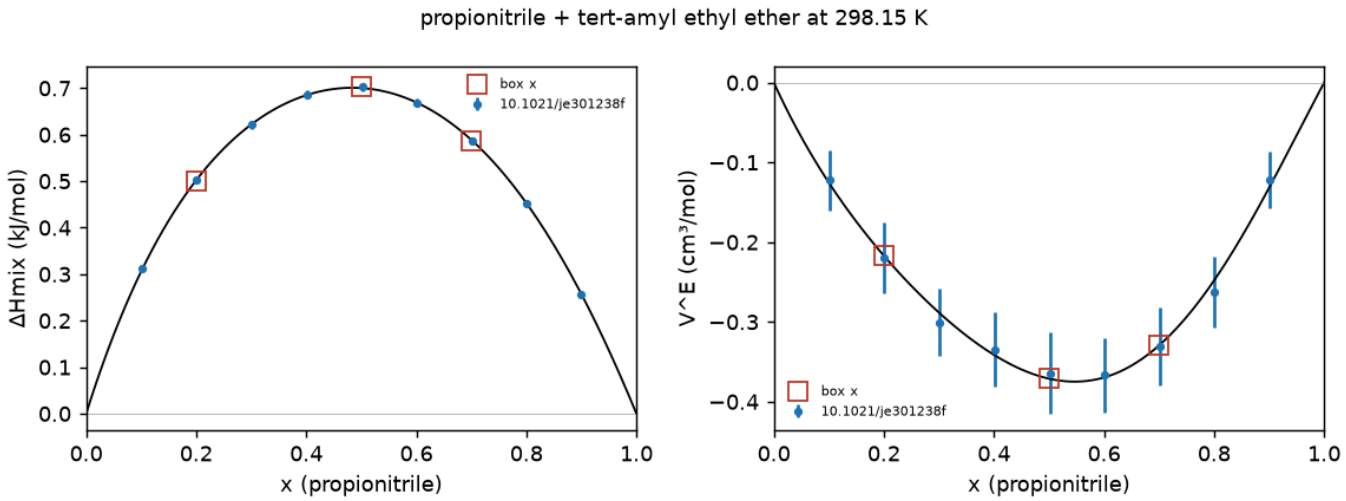


11. propionitrile + tert-amyl ethyl ether (`CCC#N` + `CCOC(C)(C)CC`)

- **Interaction:** nitrile dipoles diluted in an ether
- **Rationale:** Only nitrile system; both properties from one paper at identical compositions (data at 298.4 K). Neither component in ml-peg.

- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 9 points from 10.1021/je301238f (9)
- **Density data:** 9 points from 10.1021/je301238f (9)
- **Single-phase evidence:** ΔHmix measured at 9 compositions spanning x = 0.10–0.90; straight-line fit reduced χ^2 = 398 (a two-phase region would give ≈ 1).

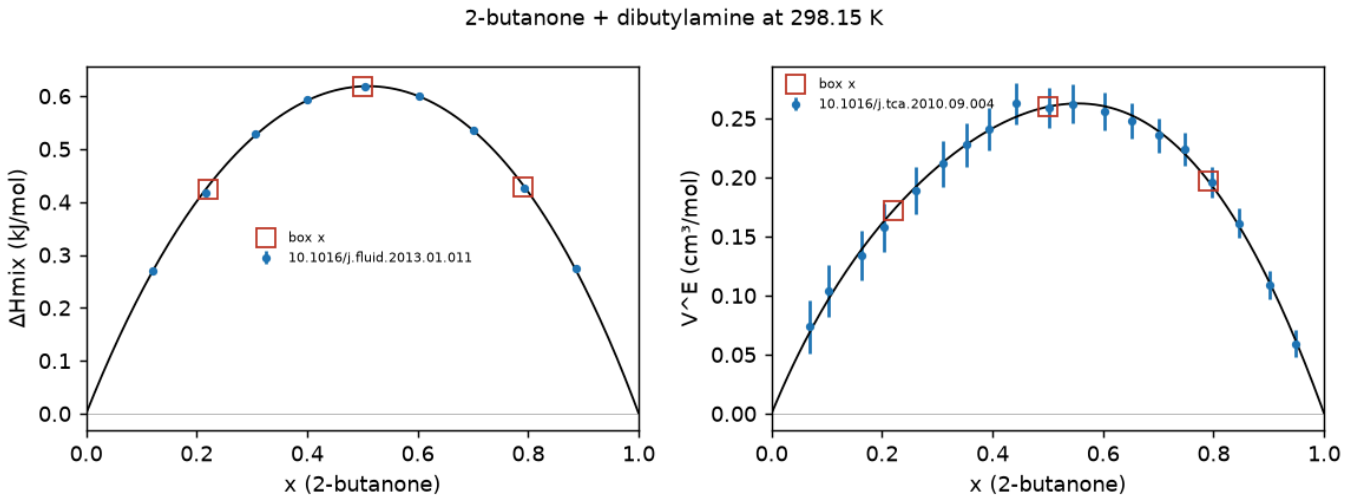
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.200	20 / 80	2100	28.3	0.501 ± 0.0069	0.7638 ± 0.00025	-0.217
0.500	50 / 50	1650	26.4	0.703 ± 0.0095	0.7683 ± 0.00035	-0.371
0.700	70 / 30	1350	25.1	0.588 ± 0.0080	0.7715 ± 0.00040	-0.330



12. 2-butanone + dibutylamine (`CCC(C)=O` + `CCCCNCCCC`)

- **Interaction:** ketone with a secondary amine
- **Rationale:** Non-alcohol endothermic system, replacing methyl pentanoate + methanol.
- **Miscibility:** Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 9 points from 10.1016/j.fluid.2013.01.011 (9)
- **Density data:** 19 points from 10.1016/j.tca.2010.09.004 (19)
- **Single-phase evidence:** ΔHmix measured at 9 compositions spanning x = 0.12–0.89; straight-line fit reduced χ^2 = 324 (a two-phase region would give ≈ 1).

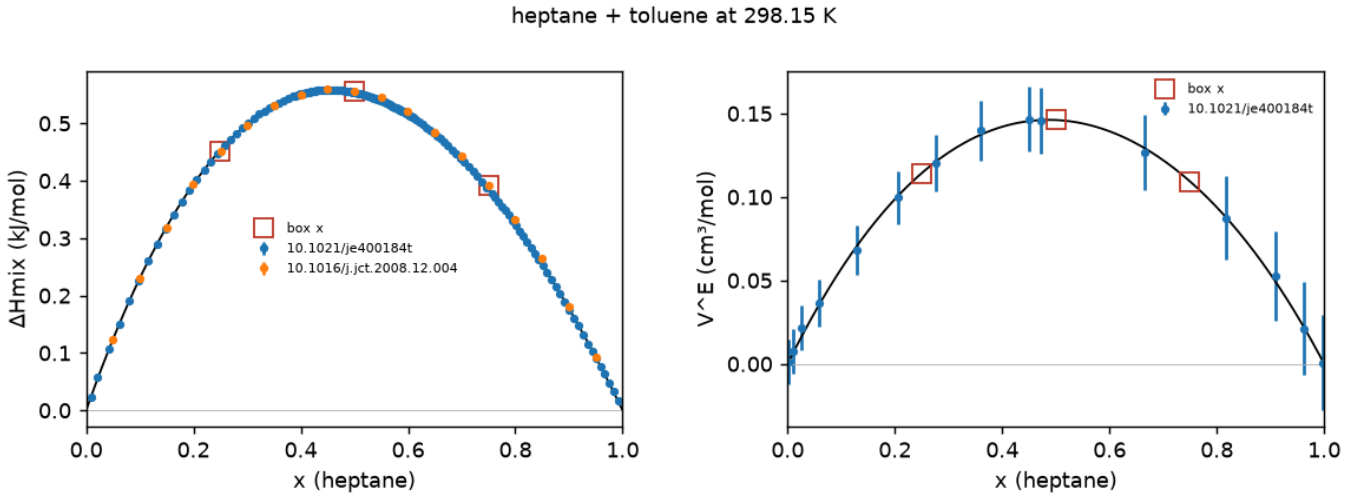
x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.220	22 / 78	2470	29.4	0.423 ± 0.0045	0.7604 ± 0.00010	0.172
0.500	50 / 50	2050	27.9	0.619 ± 0.0065	0.7693 ± 0.00010	0.260
0.790	79 / 21	1615	26.1	0.428 ± 0.0045	0.7837 ± 0.00010	0.197



13. heptane + toluene (`CCCCCCC` + `Cc1ccccc1`)

- **Interaction:** dispersion and π control (no polar groups)
- **Rationale:** Non-polar control that separates dispersion/D3 errors from hydrogen-bond errors. Sage handles it well (MAE 0.14 kJ/mol).
- **Miscibility:** Fully miscible hydrocarbons. Single liquid phase across the whole composition range in the source papers (see the single-phase evidence below).
- **ΔHmix data:** 187 points from 10.1021/je400184t (168), 10.1016/j.jct.2008.12.004 (19)
- **Density data:** 15 points from 10.1021/je400184t (15)
- **Single-phase evidence:** ΔHmix measured at 187 compositions spanning x = 0.01–0.99; straight-line fit reduced $\chi^2 = 514$ (a two-phase region would give ≈ 1).
- **Density points not used for V^E** (σ > 0.002 g/mL): 10.1016/j.jct.2005.10.011 (10)

x box	n_a / n_b	atoms	edge (Å)	ΔHmix ± σ (kJ/mol)	ρ ± σ (g/mL)	V^E (cm³/mol)
0.250	25 / 75	1700	26.9	0.451 ± 0.0030	0.8039 ± 0.00011	0.114
0.500	50 / 50	1900	27.6	0.556 ± 0.0036	0.7554 ± 0.00012	0.146
0.750	75 / 25	2100	28.4	0.391 ± 0.0026	0.7145 ± 0.00013	0.109



Known gaps

- **No N-H (secondary) amide.** Systems with N-methylformamide, N-methylacetamide, acetamide or formamide in the Sage sets:
  - `CNC=O` + `Cc1ccccc1N`: 6 points (0 with uncertainties) from sage-2.0-train
  - `C1CCOCC1` + `CNC=O`: 6 points (0 with uncertainties) from sage-2.0-train
  - `CNC=O` + `Cc1ccccc1Cl`: 6 points (0 with uncertainties) from sage-2.0-train

  The Sage 2.0 set drops their uncertainties and has enthalpies only at 308.15 K, but the ThermoML records carry full uncertainties (and the standard-InChI conversion would otherwise turn N-methylformamide into `CN=CO`; `thermoml.py` repairs amide tautomers). See candidate additions.
- **No amide-water system.** 2-pyrrolidone + water was dropped (see below).
- **Selection bias.** Every system comes from the Sage training/validation pools, so the set reflects ThermoML coverage and Sage's curation choices rather than a designed spread.
- **Endothermic range is capped** at RT/2 by the two-phase screen.

Candidate additions (pending decision)

- **N-methylformamide + o-toluidine (308.15 K)**: Closes the N-H amide gap. ThermoML (10.1016/j.tca.2011.06.020) has 18 enthalpy points (-0.41 / -0.76 / -0.61 kJ/mol at x = 0.22 / 0.52 / 0.77) and 18 densities at 308.15 K, all with uncertainties, plus same-paper pure densities; strongly single-phase. Costs 5 extra boxes at 308.15 K (3 mixtures plus N-methylformamide and o-toluidine pure boxes) and needs our own

Sage replicates, as there are no published Sage errors. Single research group; worth a cross-check against an independent source.
- **N-methylformamide + tetrahydropyran (308.15 K)**: Same paper, +0.32 kJ/mol at equimolar: below the 0.4 kJ/mol signal floor.
- **N-methylformamide + 2-chlorotoluene (308.15 K)**: ThermoML 10.1016/j.jct.2013.12.006, +0.49 kJ/mol at equimolar, 18 points of each property with uncertainties. Same research group.

Dropped systems

- **2-(dimethylamino)ethanol + water**: Sage repeat SD 0.126 kJ/mol (above the 0.10 limit).
- **2-pyrrolidone + water**: Viscous lactam that melts at ~25 °C. The Sage data use the imidic acid tautomer (OC1=NCCC1), so there is no valid Sage repeat SD, and the experimental enthalpy uncertainties are a flat 25 % placeholder.
- **piperidine + water**: Dropped after the Sage reference run (2026-10-20). Nitrogen inversion is frozen in Sage (no axial/equatorial N-H interconversion in 2.2 ns, or in a 500 K anneal), and the real barrier (~25 kJ/mol) suggests ns lifetimes for MLIPs too, so ΔHmix depended on the starting N-H orientation by ~0.3 kJ/mol. It also showed the largest box-size gap against the published 1000-molecule Sage run (-0.35 kJ/mol at x = 0.25) and is microheterogeneous for x between 0.08 and 0.5 (arXiv:2203.01224).
- **ethanol + water**: Already covered by the ethanol_water_density benchmark.
- **pyrrole + benzene**: Smallest signal and Sage already reproduces it; kept as a reserve.
- **isooctane + 1-propanol**: Replaced by 1-propanol + benzene (adds O-H···π, single-paper data).
- **methyl pentanoate + methanol**: Replaced to reduce the number of alcohol systems.
- **NMP + 1,4-dichlorobutane**: Replaced by NMP + 1,2-dichloroethane to widen the exothermic range.

Reserves

- pyrrole + benzene
- cyclohexane + benzene
- methyl pentanoate + methanol
- NMP + 1,4-dichlorobutane
- isooctane + 1-propanol
- propyl acetate + 1,2-dichloroethane

Reproduce

```
python -m \
    ml_peg.calcs.molecular_dynamics.binary_mixtures.generation.select_systems \
    --output-dir binary_mixtures_selection \
    [--fast-set dataset-fast-converging.csv]
```